

EXPERIMENT- 01

Python Code:

```
01.py > ...
1  text = input("Enter the text: ")
2  k = int(input("Enter shift value (1 to 25): "))
3
4  if 1 <= k <= 25:
5      encrypted_text = ""
6      for char in text:
7          if char.isalpha():
8              if char.isupper():
9                  encrypted_text += chr((ord(char) - 65 + k) % 26 + 65)
10             else:
11                 encrypted_text += chr((ord(char) - 97 + k) % 26 + 97)
12             else:
13                 encrypted_text += char
14
15         print("Encrypted Text:", encrypted_text)
16
17         decrypted_text = ""
18         for char in encrypted_text:
19             if char.isalpha():
20                 if char.isupper():
21                     decrypted_text += chr((ord(char) - 65 - k) % 26 + 65)
22                 else:
23                     decrypted_text += chr((ord(char) - 97 - k) % 26 + 97)
24                 else:
25                     decrypted_text += char
26
27         print("Decrypted Text:", decrypted_text)
28
29     else:
30         print("Invalid shift value! The value of k must be between 1 and 25.")
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\01.PY"
Enter the text: Hello World!
Enter shift value (1 to 25): 3
Encrypted Text: KhooR Zruog!
Decrypted Text: Hello World!
PS D:\College\Crypto\Experiments> █
```